

Class 17 Finish group material

Goals

1. Work with Cayley tables.
2. Compute order of elements.
3. Determine center and centralizers.

Definition:

G is an Abelian group means $ab = ba$ for all elements a, b in G .

Examples of Abelian groups

Examples of nonabelian groups

More definitions:

The centralizer of an element g in group G is

The center of a group is

Use the Cayley table to answer the questions below.

	1	A	B	C	D	E	F	G
1	1	A	B	C	D	E	F	G
A	A	B	C	1	G	F	D	E
B	B	C	1	A	E	D	G	F
C	C	1	A	B	F	G	E	D
D	D	F	E	G	1	B	A	C
E	E	G	D	F	B	1	C	A
F	F	E	G	D	C	A	1	B
G	G	D	F	E	A	C	B	1

1. What is the identity element?

2. What is A^{-1} ?

3. What is the order of the group?

4. What is the order of the element C?

4. What is the center of this group?

5. what is the centralizer of D?

Compare the Cayley tables for $\mathbf{Z}_6$ and for $\mathbf{U}(7)$.

z_k	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

$u(7)$	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	4	6	1	3	5
3	3	6	2	5	1	4
4	4	1	5	2	6	3
5	5	3	1	6	4	2
6	6	5	4	3	2	1

Find the order of each element in $\mathbf{Z}_6$ and $\mathbf{U}(7)$

n_L	0	1	2	3	4	5
order						

$u(7)$	1	2	3	4	5	6
order						

A blank coordinate system with a horizontal x-axis and a vertical y-axis. The y-axis is labeled with the handwritten expression $u(7)$ at the top.

Rearrange elements
in $U(7)$ and
make the Cayley table

7. (6 pts) Shown below is a blank Cayley table for the group D_4 . Write the elements in cells labeled a and b as a product of elements in D_4 . Then draw diagrams to determine exactly which element should be in the cell. You must show your diagrams to receive full credit.

a =

Diagram for a:

b =

Diagram for b:

	R_0	R_{90}	R_{180}	R_{270}	H	V	D	D'
R_0								
R_{90}								
R_{180}								
R_{270}								
H								
V		a						
D								b
D'								



H



V



D



D'

